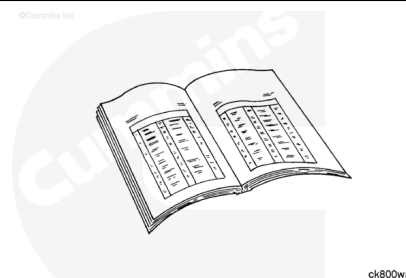


Trainings ▾

Preparatory Steps

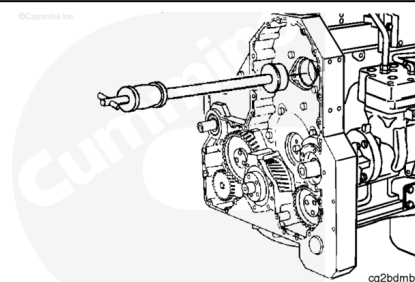
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)
- Remove the cam followers. Refer to Procedure 004-001 in Section 4. (</qs3/pubsys2/xml/en/procedures/35/35-004-001-tr.html>)
- Remove the camshaft. Refer to Procedure 001-008 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html>)



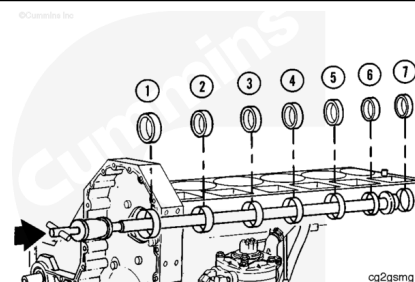
Remove

Use camshaft bushing driver, Part Number 3376637 and Part Number 3376636, or hydraulic cam bushing tool, Part Number 3823642, and hydraulic actuator kit, Part Number 3823621, to remove the camshaft bushings.

Remove bushing number one first.

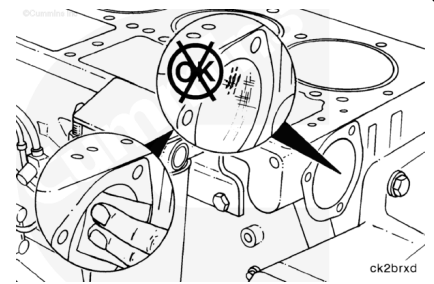


Remove the remaining six bushings in order from front to rear.



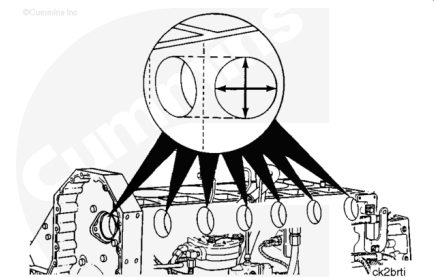
Clean and Inspect for Reuse

Use Scotch-Brite® 7448, Part Number 3823258 or equivalent, to remove burrs and to clean the bushing bores.



Measure the cylinder block camshaft bore inside diameter.

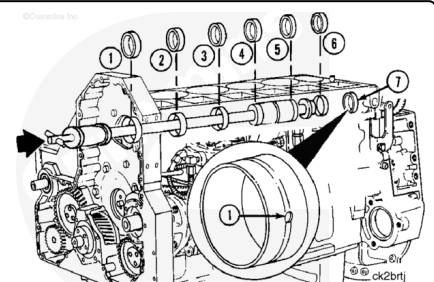
Cylinder Block Camshaft Bore I.D.		
mm		in
76.987	MIN	3.0310
77.040	MAX	3.0331



Install

Starting with Number 7 bushing, install the camshaft bushings in order from rear to front.

Note : The oil hole (1) in the bushing must be at the 3 o'clock location when viewed from the front of the engine.

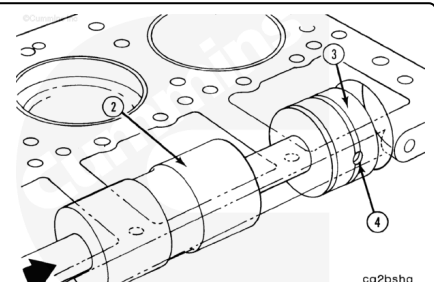


This step is for installing the Number 7 bushing.

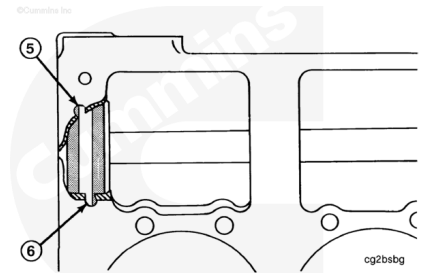
Install guide (2), Part Number 3376069, between the number six and number five bores.

Install a new bushing (3) on driver, Part Number 3376070, with the oil hole (4) at the 3 o'clock location when viewed from the front of the engine.

Drive the bushing into the bore.



The bushing oil groove must be visible in both cylinder block oil drillings (5 and 6) to prevent engine damage. The oil drillings are not in alignment with each other.



Inspect the bushing oil groove alignment with the two oil drillings in the block as the bushing is installed.

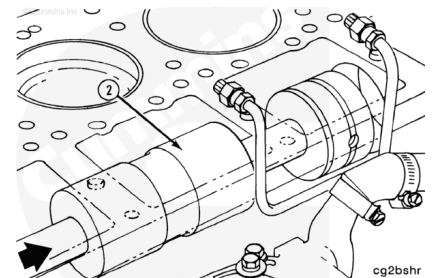
This step is for installing the Number 6 through the Number 2 bushing.

Use the same procedure to install the Number 6 through Number 2 bushings.

Install the guide (2) in the correct location.

Install the bushing on the driver with the oil hole at the 3 o'clock location when viewed from the front of the engine.

Drive in the bushing.

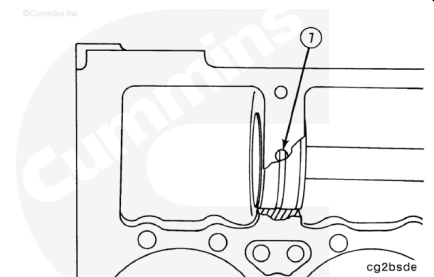


⚠ CAUTION ⚠

The oil groove must be visible through the cam follower stud holes (7). If the oil groove is not aligned, engine damage will occur.

Remove the cam follower studs. Refer to Procedure 004-001 in Section 4. (</qs3/pubsys2/xml/en/procedures/35/35-004-001-tr.html>)

Inspect the location of the camshaft bushing oil groove.

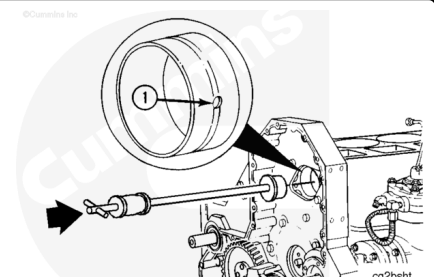


This step is for installing the Number 1 bushing.

Install a new bushing on driver, Part Number 3376070, with the oil hole (1) at the 3 o'clock location when viewed from the front of the engine.

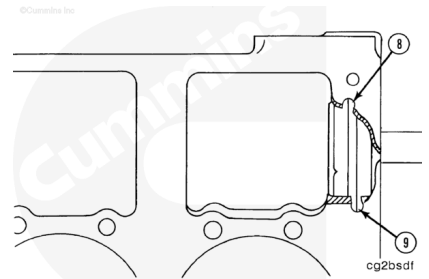
Note : The guide is not used to install this bushing.

Drive the bushing into the bore.



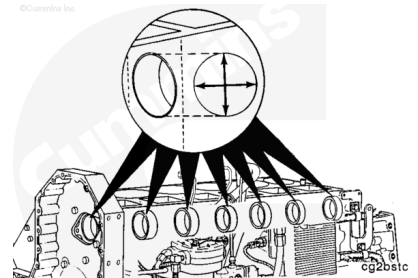
⚠ CAUTION ⚠

The bushing oil groove must be visible in both block oil drillings (8 and 9) to prevent engine damage. The block oil drillings are not in alignment with each other.



Measure the camshaft bushing-installed inside diameter.

Camshaft Bushing I.D. Installed		
mm		in
72.078	MIN	2.8377
72.142	MAX	2.8402



Finishing Steps

- Install the camshaft. Refer to Procedure 001-008 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html>)
- Install the cam follower studs and cam followers. Refer to Procedure 004-001 in Section 4. (</qs3/pubsys2/xml/en/procedures/35/35-004-001-tr.html>)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)
- Install the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Operate the engine to normal operating temperature and check for leaks.



